Next-generation Bayesian local robustness

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Berkeley Neyman Seminar
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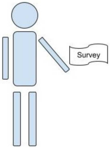
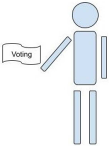
The basic problem

We have a survey population, for whom we observe:

- Covariates x (e.g. race, gender, zip code, age, education level)
- Responses y (e.g. A binary response to “do you support candidate Z”)

We want the average response in a target population, in which we observe only covariates.





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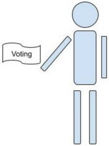

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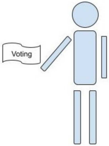

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This talk will study "Multilevel regression and poststratification" estimators:

- Set $\mathbb{E}\left[y|x,\theta\right] = m(\theta^\intercal x)$ and prior $\mathcal{P}(\theta)$ (e.g. Hierarchical logistic regression)
- Estimate the posterior $\mathcal{P}(\theta|\mathrm{Data})$ with Markov Chain Monte Carlo (MCMC)
- Take $\hat{\mu}^{\mathrm{MrP}}(Y_{\mathcal{S}}) = \frac{1}{N_T}\sum_{j=1}^{N_T} \hat{y}_j$ where $\hat{y}_j = \mathbb{E}_{\mathcal{P}(\theta|\mathrm{Data})}\left[y|x_j\right]$.

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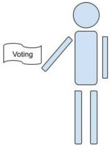

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Problem: MCMC is time consuming and the regression model is surely misspecified.
Goal: Meaningful regression specification checks with frequentist uncertainty.

Result preview

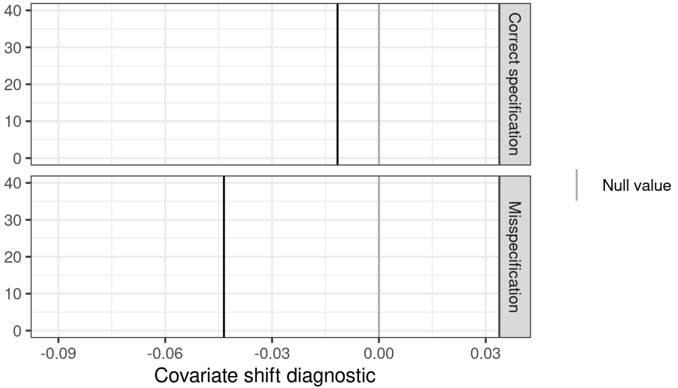

Figure 1: A covariate shift diagnostic

Here is our check for two models on the same simulated data: one correctly specified and one misspecified. The misspecied model is more extreme.

Problem: What is “large enough” to be real evidence of misspecification?
Possible answer: Frequentist confidence intervals via the parametric bootstrap.

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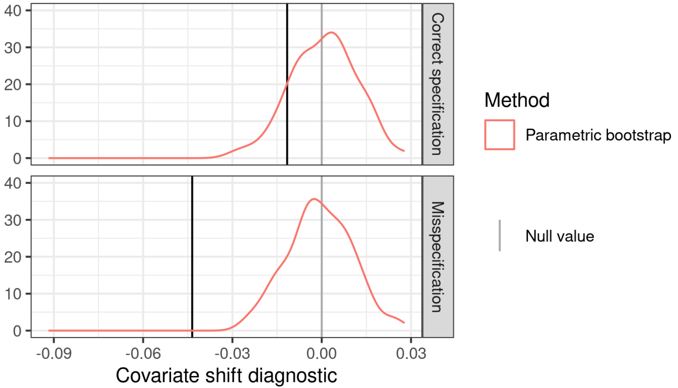

Figure 1: A covariate shift diagnostic

The frequentist interval based on the parametric bootstrap rejects the misspecified model but not the correctly specified model.

Problem: Expensive: each bootstrap sample required a new MCMC run.
Possible answer: Infintesimal jackknife confidence intervals? [Giordano and Broderick, 2026]

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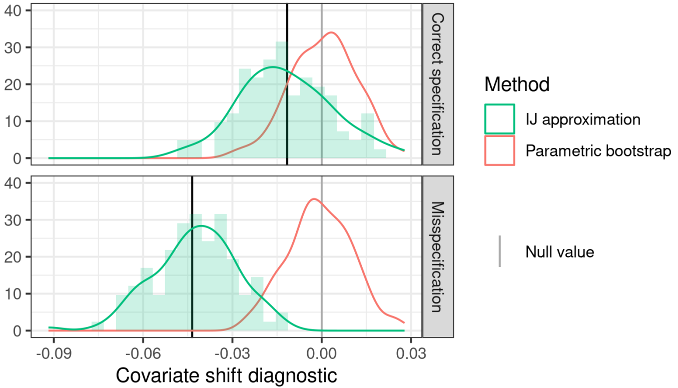

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IJ confidence intervals are fast and often work well for posterior means.
But they are not accurate for this more complex diagnostic (high Monte Carlo error).

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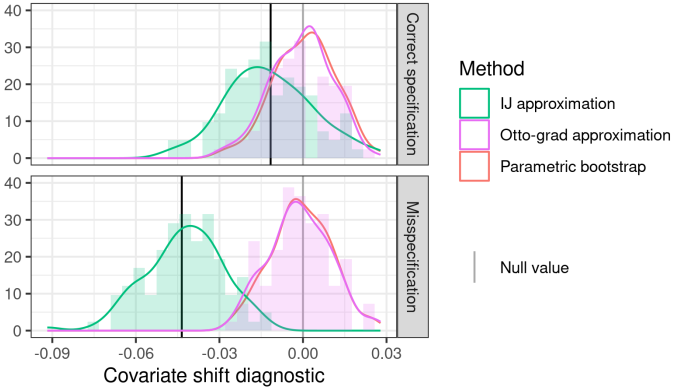

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Answer: Our flow-based local sensitivity measures ("Otto grad") produce good approximations to the bootstrap at negligible computational expense.

Outline

- Bayesian local regression specification checks
  - (With Alice Cima, Erin Hartman, Jared Miller, Avi Feller)
- Frequentist variability of Bayesian posteriors with the inifinitesimal jackknife
  - (With Tamara Broderick)
- Flow-based Bayesian local robustness: “Otto grad”
  - (With Lucas Schwengber)

Bayesian local regression specification checks

Local specification checks

$$\hat{\mu}^{\mathrm{MrP}}(Y_{\mathcal{S}}) = \frac{1}{N_T}\sum_{j=1}^{N_T} \hat{y}_j \text{ where } \hat{y}_j = \underbrace{\mathbb{E}_{\mathcal{P}(\theta|\mathrm{Data})}\left[m(\theta^\intercal x_j)\right]}_{\text{Want to check without re–running MCMC}}.$$

This model of *survey data* is definitely misspecified.

Why do we care about correct specification? We want to use $\mathcal{P}_S(x)$ to learn about $\mathcal{P}_T(x)$.

Correct specification $\Rightarrow$ Regression estimates are invariant to $\mathcal{P}_S(x)$

Idea: Check invariance to $\mathcal{P}_S(x)$ directly.

Let $\mathcal{P}(x, y)$ denote a generic joint density of x, y.
Let $\hat{\mathcal{P}}_S$ the empirical distribution on the survey data.

1. Write our estimand as a functional $\mu(\mathcal{P})$ that takes in a density and returns a MrP estimate
2. Define a covariate density perturbation $w(x)$
3. Define $\mathcal{P}_\epsilon(x, y) = \mathcal{P}(x, y) + \epsilon w(x)\mathcal{P}(y|x) = (\mathcal{P}(x) + \epsilon w(x))\, \mathcal{P}(y|x)$
4. Compute the derivative $\Delta(w, \mathcal{P}) = \partial\mu(\mathcal{P}_\epsilon)/\partial\epsilon$
5. Plug the empirical distribution into $\Delta(w, \hat{\mathcal{P}}_S)$ and check whether ≈ 0

Bayesian sensitivity analysis

Consider a family of distributions $\mathcal{P}^{\epsilon}(\theta) \propto \mathcal{P}(\theta)\exp(\epsilon h(\theta))$ and a test function $\phi(\theta)$. Then, by interchange of integration and differentiation,

$$\left.\frac{\partial \mathbb{E}_{\mathcal{P}^{\epsilon}(\theta)}\left[\phi(\theta)\right]}{\partial \epsilon}\right|_{\epsilon=0} = \mathrm{Cov}_{\mathcal{P}(\theta)}\left(\phi(\theta), h(\theta)\right) .$$

Let the regression log likelihood be $\ell(y|x,\theta)$, so that

$$\mathcal{P}(\theta|\mathrm{Data}) \propto \exp\left(\sum_{i=1}^{N_S} \ell(y_i|x_i,\theta)\right) \text{ and } \mathcal{P}^{\epsilon}(\theta|\mathrm{Data}) \propto \mathcal{P}(\theta|\mathrm{Data}) \exp\left(\sum_{i=1}^{N_S} \epsilon w(x_i)\ell(y_i|x_i,\theta)\right)$$

- $\mathcal{P}(\theta) = \mathcal{P}(\theta|\mathrm{Data})$
- $\phi(\theta) = \frac{1}{N_T}\sum_{j=1}^{N_T} m(\theta^{\intercal}x_j)$ (so that $\hat{\mu}^{\mathrm{MrP}}(Y_{\mathcal{S}}) = \mathbb{E}_{\mathcal{P}(\theta|\mathrm{Data})}\left[\phi(\theta)\right]$)
- $h(\theta) = \sum_{i=1}^{N_S} w(x_i)\ell(y_i|x_i,\theta) \Rightarrow$

Our diagnostic is then given by

$$\Delta(w, \hat{\mathcal{P}}_S) := \left.\frac{\partial \hat{\mu}^{\mathrm{MrP}}(Y_{\mathcal{S}})}{\partial \epsilon}\right|_{\epsilon=0} = \mathrm{Cov}_{\mathcal{P}(\theta|\mathrm{Data})}\left(\frac{1}{N_T}\sum_{j=1}^{N_T} m(\theta^{\intercal}x_j), \sum_{i=1}^{N_S} w(x_i)\ell(y_i|x_i,\theta)\right)$$

We expect $\Delta(w, \hat{\mathcal{P}}_S)$ to be close to zero when the model is correctly specified and posterior estimation is accurate.

Simulation

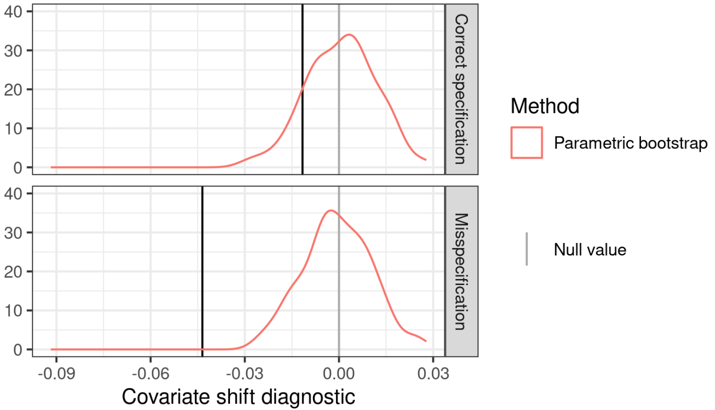

Figure 2: A covariate shift diagnostic

Simulation data: Logistic regression with five active regressors, each with covariate shift.

$$\text{Model 0 (correct)} : \mathtt{y} \sim \text{Logistic}(\mathtt{1\ +\ X1\ +\ X2\ +\ X3\ +\ X4\ +\ X5})$$

$$\text{Model 1 (misspecified)} : \mathtt{y} \sim \text{Logistic}(\mathtt{1\ +\ X3\ +\ X4\ +\ X5})$$

Compute diagnostic with $w(x) = \mathbb{I}\,(\mathtt{X2} > 0.878)$

Frequentist variance with the infinitesimal jackknife

Why frequentist variance?

For the remainder of the talk, focus on approximating the *frequentist* sampling distribution under $x_i, y_i \overset{iid}{\sim} \mathcal{P}_S(x, y)$ of quantities like our diagnostic:

$$\Delta = \mathrm{Cov}_{\mathcal{P}(\theta|\text{Data})}\left(\frac{1}{N_T}\sum_{j=1}^{N_T} m(\theta^\intercal x_j), \sum_{i=1}^{N_S} w(x_i)\ell(y_i|x_i, \theta)\right)$$

No need to commit to our covariate shift diagnostic to care!

- Frequentist variablity of *any* model checking diagnostic is interesting
- Frequentist variablity remains meaningful under model misspecification (Huggins and Miller, IJ papers)

We will form *local approximations* computable only from *the original posterior*.

We will compare with computationally expensive bootstrap:

- Compute Δ on the original dataset
- For $B = 1, \ldots, 100$:
  1. Sample $(x_1^*, y_1^*), \ldots (x_{N_S}^*, y_{N_S}^*)$ (parametric or nonparametric bootstrap)
  2. Run MCMC to estimate $\mathcal{P}(\theta|(x_1^*, y_1^*), \ldots (x_{N_S}^*, y_{N_S}^*))$
  3. Use the draws to estimate Δ_B^*
- Compare Δ with quantiles of $\Delta_B^* - \Delta$

Data re-weighting.

Augment the problem with *data weights* $w_1, \ldots, w_N$. We can write $\mathbb{E}_{\mathcal{P}(\theta|X;w)}\left[f(\theta)\right]$.

$$\ell_n(\theta) := \log \mathcal{P}(x_n|\theta) \qquad\qquad \log \mathcal{P}(X|\theta; w) = \sum_{n=1}^{N} w_n \ell_n(\theta)$$

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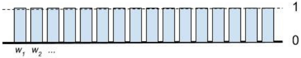

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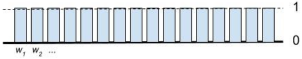

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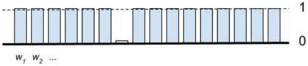

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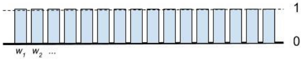

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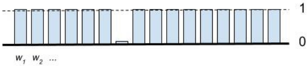

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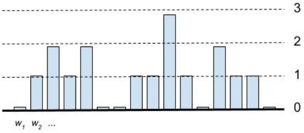

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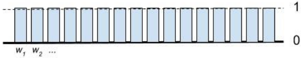

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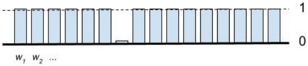

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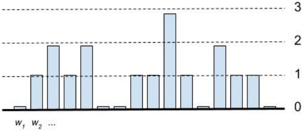

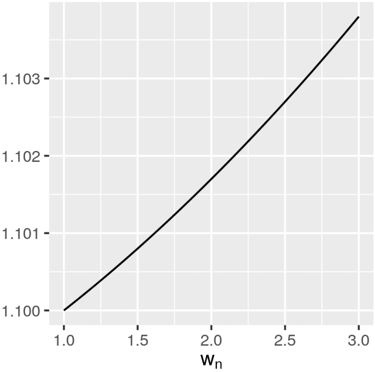

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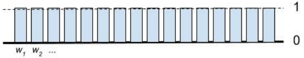

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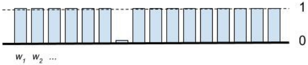

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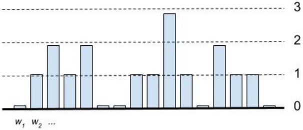

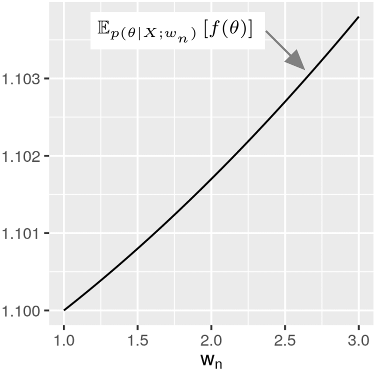

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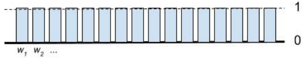

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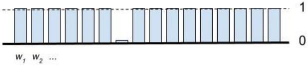

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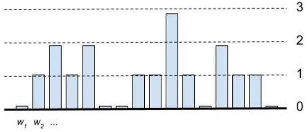

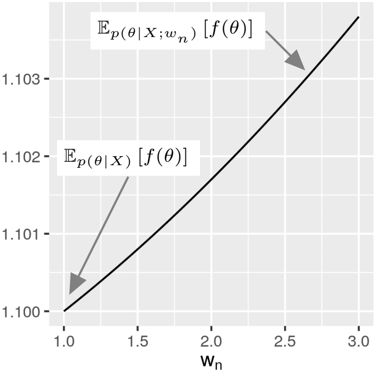

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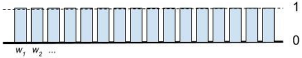

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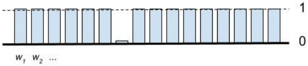

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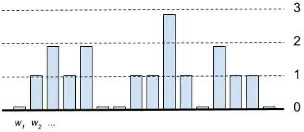

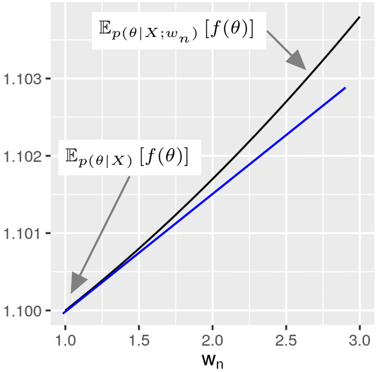

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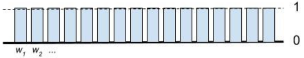

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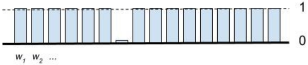

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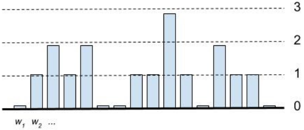

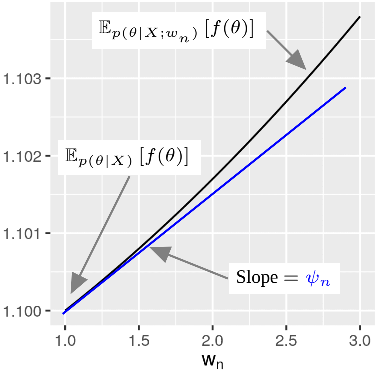

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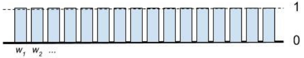

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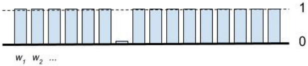
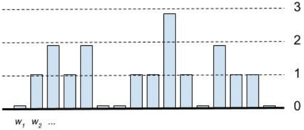
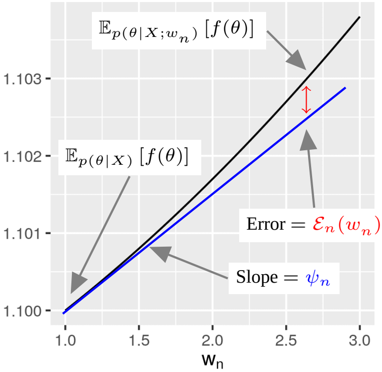


Bootstrap weights:



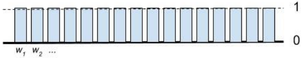


Data re-weighting.

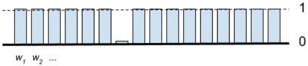
Augment the problem with *data weights* $w_1, \dots, w_N$. We can write $\mathbb{E}_{\mathcal{P}(\theta|X;w)}[f(\theta)]$.

$$\ell_n(\theta) := \log \mathcal{P}(x_n|\theta) \qquad \log \mathcal{P}(X|\theta; w) = \sum_{n=1}^{N} w_n \ell_n(\theta)$$

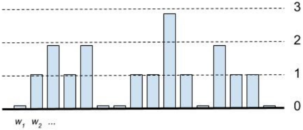
Original weights:

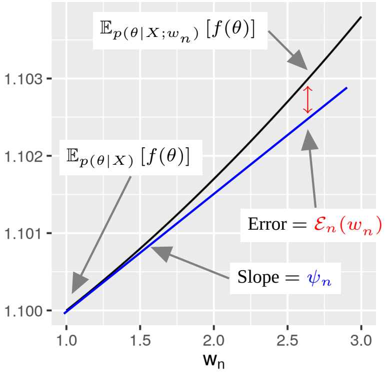


Leave-one-out weights:



Bootstrap weights:





The re-scaled slope $N\psi_n$ is known as the "influence function" at data point x_n.

$$\mathbb{E}_{p(\theta|X;w)}[f(\theta)] - \mathbb{E}_{p(\theta|X)}[f(\theta)] = \sum_{n=1}^{N} \psi_n (w_n - 1) + \mathcal{E}_n(w)$$

Data re-weighting.

How can we use the approximation?

Assume the slope is computable and error is small.

$$\mathbb{E}_{p(\theta|X;w)}\left[f(\theta)\right] - \mathbb{E}_{p(\theta|X)}\left[f(\theta)\right] = \sum_{n=1}^{N} \psi_n (w_n - 1) + \mathcal{E}_n(w)$$

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Bootstrap. Draw bootstrap weights $w \sim \mathcal{P}(w) = \text{Multinomial}(N, N^{-1})$.

$$\text{Bootstrap variance} = \text{Var}_{\mathcal{P}(w)}\left(\mathbb{E}_{p(\theta|X;w)}\left[f(\theta)\right]\right)$$

Data re-weighting.

How can we use the approximation?

Assume the slope is computable and error is small.

$$\mathbb{E}_{p(\theta|X;w)}\left[f(\theta)\right] - \mathbb{E}_{p(\theta|X)}\left[f(\theta)\right] = \sum_{n=1}^{N} \psi_n (w_n - 1) + \mathcal{E}_n(w)$$

Bootstrap. Draw bootstrap weights $w \sim \mathcal{P}(w) = \text{Multinomial}(N, N^{-1})$.

$$\begin{aligned}
\text{Bootstrap variance} &= \text{Var}_{\mathcal{P}(w)}\left(\mathbb{E}_{p(\theta|X;w)}\left[f(\theta)\right]\right) \\
&= \text{Var}_{\mathcal{P}(w)}\left(\sum_{n=1}^{N} \psi_n (w_n - 1) + \mathcal{E}_n(w)\right) \\
&= \frac{1}{N^2} \sum_{n=1}^{N} \left(\psi_n - \overline{\psi}\right)^2 + \text{Term involving } \mathcal{E}_n(w) \text{ for } n = 1, \ldots, N \\
&\approx \underbrace{\frac{1}{N}\left(\frac{1}{N}\sum_{n=1}^{N}\left(\psi_n - \overline{\psi}\right)^2\right)}_{\text{“Infinitesimal jackknife variance estimate”}}
\end{aligned}$$

Bayesian von–Mises Expansion

Define the "generalized posterior" functional [Giordano and Broderick, 2026]:

$$T(\mathcal{P}, N) := \frac{\int g(\theta) \exp\left(N \int \ell(y|x,\theta)\mathcal{P}(x,y)dxdy\right) \pi(\theta)d\theta}{\int \exp\left(N \int \ell(y|x,\theta)\mathcal{P}(x,y)dxdy\right) \pi(\theta)d\theta}.$$

Recall that $\mathbb{E}_{\mathcal{P}(\theta|\mathrm{Data})}\left[g(\theta)\right] = T(\hat{\mathcal{P}}_S, N_S)$

To derive frequentist variance estimates, study the *von Mises expansion*.
Let $\mathcal{P}_t = t\hat{\mathcal{P}}_S + (1-t)\mathcal{P}_S$. Then form the series expansion in t:

$$\begin{aligned}
\sqrt{N_S}\left(\mathbb{E}_{\mathcal{P}(\theta|\mathrm{Data})}\left[g(\theta)\right] - T(\mathcal{P}_S, N_S)\right) &= \sqrt{N_S}\left(T(\hat{\mathcal{P}}_S, N_S) - T(\mathcal{P}_S, N_S)\right) \\
&= \sqrt{N_S}\left.\frac{\partial T(\mathcal{P}_t, N_S)}{\partial t}\right|_{t=0}(1-0) + \mathcal{E}(\tilde{t}) \\
&= \underbrace{\sqrt{N_S}\sum_{n=1}^{N}(\psi_i - \overline{\psi})}_{\text{Infinitesimal jackknife estimator}} + o_p(1) + \mathcal{E}(\tilde{t}).
\end{aligned}$$

where

$$\psi_i := N_S \mathrm{Cov}_{\mathcal{P}(\theta|\mathrm{Data})}\left(g(\theta), \ell(y_i|x_i)\right) \text{ and } \tilde{t} \in [0,1].$$

Bayesian von–Mises Expansion Results

$$\sqrt{N_S}\left(\mathbb{E}_{\mathcal{P}(\theta|\text{Data})}\left[g(\theta)\right] - T(\mathcal{P}_S, N_S)\right) = \sqrt{N_S}\sum_{n=1}^{N}(\psi_i - \overline{\psi}) + o_p(1) \qquad + \mathcal{E}(\tilde{t}).$$

- Assume (sketch): A well–behaved MAP *maximum a posteriori* estimator $\hat{\theta}$ exists.
  - The dimension of θ is fixed as $N \to \infty$
  - The expected log likelihood has a strict maximum at θ_0
  - The observed log likelihood statisfies $\hat{\theta} \to \theta_0$
  - The expected log likelihood Hessian is negative definite at θ_0
- Assume (sketch): We can apply standard asymptotics.
  - The log prior and log likelihood are four times continuously differentiable
  - The prior is proper, and a technical set of prior expectations are finite
  - The log likelihood and its derivatives are dominated by a square–integrable envelope function for all θ in a neighborhood of θ_∞.

Theorem 3 [Giordano and Broderick, 2026] (sketch):

$$\sup_{\tilde{t}\in[0,1]} \left|\mathcal{E}(\tilde{t})\right| \xrightarrow{p} 0 \quad \text{in the Bayesian von–Mises expansion.}$$

Corollary: Let $\frac{1}{N_S}\sum_{i=1}^{N_S}(\psi_i - \overline{\psi})^2 \xrightarrow{p} V$. By the CLT applied to ψ_i,

$$\sqrt{N_S}\left(\mathbb{E}_{\mathcal{P}(\theta|\text{Data})}\left[g(\theta)\right] - T(\mathcal{P}_S, N_S)\right) \xrightarrow{d} \mathcal{N}\left(0, V\right).$$

Data re-weighting.

Recall that our covariate shift diagnostic is given by

$$\Delta(w, \hat{\mathcal{P}}_S) = \mathrm{Cov}_{\mathcal{P}(\theta|\mathrm{Data})}\left(\frac{1}{N_T}\sum_{j=1}^{N_T} m(\theta^\intercal x_j), \sum_{i=1}^{N_S} w(x_i)\ell(y_i|x_i,\theta)\right) = \sum_{i=1}^{N_S}\psi_i w(x_i).$$

In other words, we are checking whether the influence function aligns with any of the regressors.

Data re-weighting.

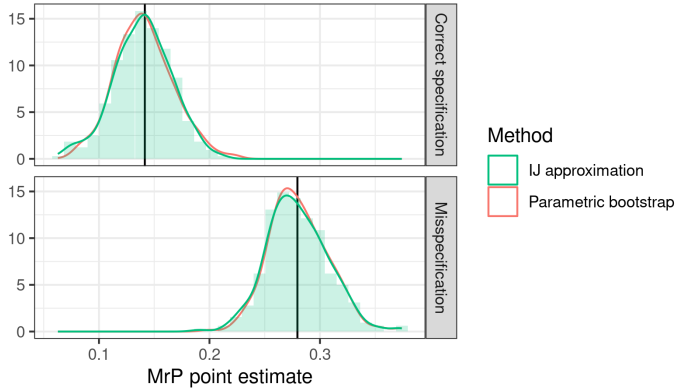

Figure 3: MrP Posterior Mean

Data re-weighting.

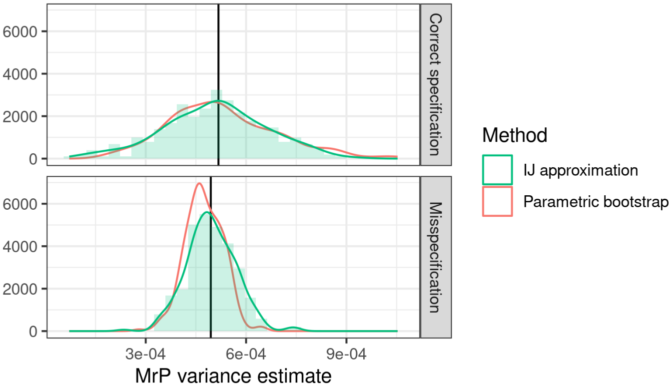

Figure 3: MrP Posterior Variance

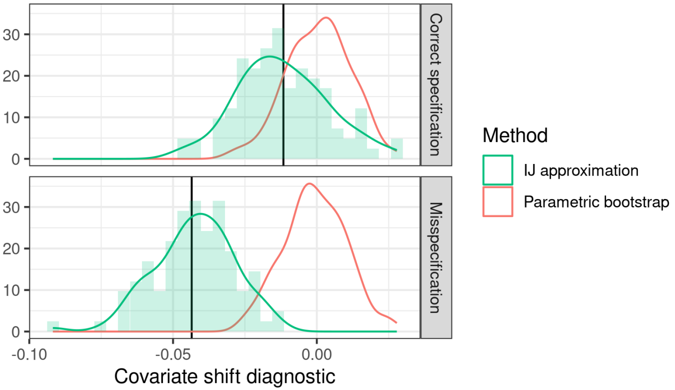

Figure 3: Covariate Shift Diagnostic

Local robustness with flows

Why flows for sensitivity analysis?

The IJ is built around linearizing posterior expectations:

$$\mathbb{E}_{\mathcal{P}^{\epsilon}(\theta)}\left[g(\theta)\right] \approx \mathbb{E}_{\mathcal{P}}\left[g(\theta)\right] + \left.\frac{\partial \mathbb{E}_{\mathcal{P}^{\epsilon}(\theta)}\left[g(\theta)\right]}{\partial \epsilon}\right|_{\epsilon=0} (\epsilon - 0),$$

where the derivative is estimated — with noise — from MCMC draws.

This has some shortcomings:

- Potential MCMC variance, not obvious how to smooth
- Linear extrapolation can lead to impossible values (e.g. negative variances)
- Complex quantities like posterior quantities are hard to linearize
- Bayesians like to work with draws, not analytic expectations

Idea: Instead of linearizing expectations, learn a flow from $\mathcal{P}$ to $\mathcal{P}^{\epsilon}$.

There's a large literatue on learning flows between distant distributions. Our problem is easier:

- Only need "local" flows
- Our perturbations have a lot of structure (e.g. bootstraps)

Flow

Consider a family of distributions $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta)\exp(\epsilon h(\theta))$ and a test function $\phi(\theta)$. Then, by interchange of integration and differentiation,

$$\left.\frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)}\left[\phi(\theta)\right]}{\partial \epsilon}\right|_{\epsilon=0} = \mathrm{Cov}_{\mathcal{P}(\theta)}\left(\phi(\theta), h(\theta)\right) .$$

We want a flow $\theta \mapsto f^\epsilon(\theta)$ such that, for small ϵ, $f^\epsilon(\theta) \sim \mathcal{P}^\epsilon(\theta)$ and $f^0(\theta) = \theta$.

Suppose there is a scalar-valued potential function $v(\theta)$ such that

$$f^\epsilon(\theta) = \theta + \epsilon \nabla v(\theta) + O(\epsilon^2) \quad \text{as } \epsilon \to 0. \text{ Then:}$$

$$\begin{array}{ccc} \mathbb{E}_{\mathcal{P}^\epsilon(\theta)}\left[\phi(\theta)\right] & = & \mathbb{E}_{\mathcal{P}(\theta)}\left[\phi(f^\epsilon(\theta))\right] \Rightarrow \\ \left.\frac{\partial}{\partial \epsilon}\right|_{\epsilon=0} \mathbb{E}_{\mathcal{P}^\epsilon(\theta)}\left[\phi(\theta)\right] & = & \left.\frac{\partial}{\partial \epsilon}\right|_{\epsilon=0} \mathbb{E}_{\mathcal{P}(\theta)}\left[\phi(\theta + \epsilon \nabla v(\theta))\right] \\ \| & & \| \\ \mathbb{E}_{\mathcal{P}(\theta)}\left[\phi(\theta) h(\theta)\right] & = & \mathbb{E}_{\mathcal{P}(\theta)}\left[\nabla \phi(\theta)^\intercal \nabla v(\theta)\right] \end{array}$$

- When does such a $v(\theta)$ exist?
- How can we find it in practice?

When does such a $v(\theta)$ exist?

Consider a family of distributions $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta)\exp(\epsilon h(\theta))$.
Integration by parts (with compactly supported ϕ) gives:

$$\begin{aligned}
\mathbb{E}_{\mathcal{P}(\theta)}\left[\phi(\theta)h(\theta)\right] &= \mathbb{E}_{\mathcal{P}(\theta)}\left[\nabla\phi(\theta)^\intercal\nabla v(\theta)\right] \\
&= \mathbb{E}_{\mathcal{P}(\theta)}\left[\phi(\theta)\left(-\nabla v(\theta)^\intercal\nabla\log\mathcal{P}(\theta) - \Delta v(\theta)\right)\right] \\
&=: \mathbb{E}_{\mathcal{P}(\theta)}\left[\phi(\theta)\mathcal{L}_{\mathcal{P}}(v)(\theta)\right] \text{ for} \\
\mathcal{L}_{\mathcal{P}}(v) :=& -\nabla v(\theta)^\intercal\nabla\log\mathcal{P}(\theta) - \Delta v(\theta).
\end{aligned}$$

This is a weak formulation of the linear differential equation

$$\mathcal{L}_{\mathcal{P}}(v) = h.$$

Note that $\mathcal{L}_{\mathcal{P}}$ is the drift operator, generator of Langevin diffusion, trace of the Stein operator.

Theorem (Schewengber): Assume that $\log\mathcal{P}(\theta) \in C^2(\mathcal{R}^d)$ and

$$\lim_{\theta\to\infty}\left(\frac{1}{2}\|\nabla\log\mathcal{P}(\theta)\|^2 - \Delta\log\mathcal{P}(\theta)\right) = +\infty.$$

Then $\mathcal{L}_{\mathcal{P}}^{-1}$ is bounded as a map from $\dot{H}^1(p) \mapsto \dot{H}^1(p)$.

Corollary: If $\log\mathcal{P}(\theta|\text{Data})$ is sufficiently regular, $\mathbb{E}_{\mathcal{P}(\theta|\text{Data})}\left[\|\nabla h(\theta)\|_2^2\right] < \infty$, and $\left|\mathbb{E}_{\mathcal{P}(\theta|\text{Data})}\left[h(\theta)\right]\right| < \infty$, then there exists a differentiable flow potential $v(\theta)$ that locally approximates $\mathcal{P}^\epsilon(\theta|\text{Data})$.

How can we find $v(\theta)$ in practice?

Fix $h(\theta)$. Specify

- A family of real-valued test functions $\phi_n(\theta)$ for $n \in [N]$
- A family of real-valued basis functions $b_m(\theta)$ for $m \in [M]$

$$\text{Let } v(\theta; \beta) := \sum_{m=1}^{M} \beta_m b_m(\theta). \text{ Then}$$

$$\underbrace{\mathbb{E}_{\mathcal{P}(\theta)}\left[\phi_n(\theta) h(\theta)\right]}_{\text{Estimate with MCMC!}} = \sum_{m=1}^{M} \beta_m \underbrace{\mathbb{E}_{\mathcal{P}(\theta)}\left[\nabla \phi_n(\theta)^\intercal \nabla b_m(\theta)\right]}_{\text{Estimate with MCMC!}}$$

This gives N linear equations in M unknowns. Unlike standard linear regression:

- We are free to choose ϕ_n and b_m
- Both F and B are measured with noise
- We have a lot of information about the covariance of F and B

Theorem (Schewengber): The solution gives the best L2 approximation to $\|\nabla v(\theta)\|_{L^2(\mathcal{P})}$ in the span of $\phi_n(\theta)$. When $\mathcal{P}$ is Gaussian, the ordered Hermite polynomials give the best finite-rank approximation.

References

R. Giordano and T. Broderick. The Bayesian infinitesimal jackknife for variance. *arXiv preprint arXiv:2305.06466*, 2026.